

**Visual Mental Imagery is Not Evidently Separable from Episodic
Memory Recall**

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Course: Master's Thesis, 30 ECTS

Spring 2024

I dedicate this thesis to my father.

Visual Mental Imagery is Not Evidently Separable from Episodic Memory Recall

Upon forming a visual mental image of a friend's face, we have a mental experience to which we can assign a certain degree of vividness. Upon recalling a friend's face, a similar mental experience occurs to which a similar level of vividness may also apply. Do these two expressions (forming a visual mental image or recalling) underlie the same process given the perceptual nature of the content: the friend's face? To put it differently, is there a difference in the meaning of vividness between visual mental images and recollections? Traditionally, the term vividness is a property used in visual mental imagery research (Marks, 1973; McKelvie, 1995a, 1995b)¹. To measure it, Marks created the Vividness of Visual Imagery Questionnaire (VVIQ, 1973) where participants are asked to consider four scenes, over 16 items, and rate their vividness on a 5-point Likert scale. The scale ranges from *Perfectly clear and as vivid as normal vision* to *No image at all, you only "know" that you are thinking of the object*.

There is, however, a concern over what vividness of visual mental images might mean. According to Fazekas et al. (2020) vividness is a vague term that has no proper scientific definition. In fact, literature demonstrates many interpretations. Marks' original description (1973) was about liveliness (a synonym for vividness) and clarity. These two properties are repeated by McKelvie (1995b), stating that vividness is "similarity in clarity and liveliness to experiences that have occurred in perception" (p. 204). D'Angiulli and Reeves (2007) also utilise clarity but expand the definition with richness and amount of detail. Baddeley and Andrade (2000) remain more succinct: "richness of representations" (p. 128). Schwitzgebel's list (collected from further literature) is lengthier: "precision of outline, amount of detail, brightness, saliency, 'liveliness' and other factors" (2002, p. 48). As Fazekas et al. (2020) note, the problem is that vividness is explained by concepts that by themselves have no clear definitions. A further problem derives from the argument that participants may interpret their vividness experiences differently as there is no objective criterion to which vividness could be measured and compared (McKelvie, 1995a; Schwitzgebel, 2002). Two people might have a similar visual mental image of a friend's face, yet one judges its vividness to be *perfectly clear* whereas the other judges it to be just *clear*. This argument is not about whether such mental experiences are similar or not but whether people have similar or different criteria as to what different levels of vividness mean for them. As such, it becomes a difficult methodological exercise to compare two participants' vividness ratings.

The terms used interchangeably to define vividness refer to visual qualities. This indicates that a vivid visual mental image resembles visual perception. From a theoretical perspective, this claim echoes in the mental imagery debate, where two opposing views compete over the nature (or the format) of mental imagery (Pearson & Kosslyn, 2015). As per the depictive view, mental images have similar spatial structures as their corresponding external (perceptual) objects. As per the propositional view, mental images cannot be described in spatial terms for these involve higher-level propositional attitudes, for example, beliefs (as opposed to lower-level visual features, such as size or shape) that are symbolic in their nature (Pylyshyn, 1979). Empirical evidence, however, supports the depictive format of mental images (Pearson & Kosslyn, 2015). Accordingly, vividness – however ill-defined it may be – is a property of visual mental imagery.

Another aspect of visual mental imagery vividness is that it is strongly connected to memory. The VVIQ contains four familiar scenes to visualize (i.e., to form a visual mental image): the face of a friend or a relative, the rising of the sun, a shop we often go to, and a country scene. According to Tulving (1985), episodic memory is about "personally experienced events" (p. 2). In fact, the scenes selected for the VVIQ originally aimed to be as close to "personally experienced events" as possible (Marks, 1972, cited in McKelvie, 1995a). As Baddeley and Andrade (2000) contend, the more information is retrieved from long-term

¹ Not considering "broader" categories of the mind and cognition, such as vividness of perception or emotions (Fazekas et al., 2020).

memory (which includes episodic memory as well), the more vivid a visual mental image becomes. If answering the VVIQ items depends on episodic memory, how can the two theoretical constructs – episodic memory and visual mental imagery – be disentangled? An answer would be to make explicit instructions to the participants what to focus on, which is in fact the case for the VVIQ. However, an instruction on “visualizing” a friend’s face does not (and cannot) exclude “recalling” it. The two constructs – I will argue – are not easy to separate.

Initially, vividness was an attribute of visual mental imagery. Yet, it is often associated with episodic memory as well. Cooper and Ritchey (2022) state that vividness is a property used both in memory and mental imagery research. In their experiments participants were asked to rate the vividness of remembered events. The rating scale ranged from *not at all vividly* to *extremely vividly*. Note, that in addition to this task they also employed the VVIQ (which utilises at least one additional interpretation of vividness, see above). That is, vividness of episodic memory was measured on a trial basis (encoding and retrieval trials of events), whereas vividness of mental imagery was assessed by a subjective questionnaire. It remains unclear, however, whether episodic memory vividness means the same as visual mental imagery vividness, due to being employed in different task designs. Geib et al. (2017) measured vividness of recollections of previously seen images, on a scale ranging from *little or no detail* to *highly detailed*. Note, that these labels reflect one interpretation of vividness – amount of detail – that was previously linked to visual mental imagery (D’Angiulli & Reeves, 2007). The procedure of Geib et al. (2017) implies that participants would rate mental imagery vividness², yet it is unclear whether their instructions explicated this or not. In another study, participants were instructed to “close their eyes and vividly recall the cued memory” (Chadwick et al., 2010, p. 546). However, literature on vividness of mental imagery traditionally asks their participants to visualize events first with eyes closed, then with eyes open (see e.g., Marks, 1973). McKelvie (1995a) discusses that with eyes open it is possible that visual perception distracts mental imagery vividness³. It could be the case that with eyes open it is harder to focus on the visual details of a mental image due to being surrounded by external visual input. Thus, referring to eyes closed in an episodic memory vividness task implies the usage of visual mental imagery. In a related study (Lindeman et al., 2017), participants were asked to rate the vividness of recalled memory events, with a lowest rating being 1 – *No image at all* and the highest being 5 – *Perfectly clear as if I were actually seeing it*. Such rating descriptions imply visual mental imagery towards participants. However, the authors label the construct as “memory vividness” and not “mental imagery vividness”. In fact, mental imagery was not measured directly in this study. In summary, these studies show a yet unnoticed concern: while researchers make conclusions on one construct (episodic memory), their methodologies imply the usage of a secondary construct towards participants (visual mental imagery).

These concerns are becoming increasingly important with a new wave of research on those who mostly report only “knowing” about thinking scenes or objects. *Aphantasia* – the label for those who cannot voluntarily form vivid visual mental images – was coined by Zeman et al. (2015) based on people scoring low on the VVIQ. The 21 volunteering participants, who had stated no mental imagery experiences prior to the study, completed a modified version of the VVIQ (including 121 control participants) such that the rating scale utilised “lively” as opposed to “vivid”. The lowest rating remained the original (Marks, 1973), whereas the highest rating became *Perfectly clear and lively as real seeing*. Results showed that the median VVIQ score of the 21 voluntary participants was 16 (the lowest possible score, with the lowest rating on each of the 16 items of the VVIQ), whereas control participants reached a median score of 58. No imagers (total VVIQ scores = 16) were further distinguished from minimal imagers (total VVIQ scores = 17-30), although the label aphantasia covers both groups. Since 2015, a surge of studies compared aphantasia to non-aphantasia by utilising the VVIQ and setting up different cut-off points to identify aphantasia. Zeman et al. (2020) defined it as having VVIQ

² “... rated the amount of detail (vividness) in the *image* they generated” (p. 682, my emphasis).

³ Although, McKelvie did not find this a serious confounding variable.

scores between 16 and 23. Pounder et al. (2022) distinguished aphantasia (scoring below 25) from extreme cases of aphantasia (scoring only 16). In a comprehensive study (Takahashi et al., 2023) that analysed 2,871 participants – including controls – the VVIQ cut-off point for aphantasia was 32 that included participants both with ($n = 46$) and without ($n = 59$) self-identification with aphantasia (i.e., self-identification prior to the study). However, the authors also found participants who had self-identified with aphantasia but scored above 33 ($n = 301$). The absence of an agreed VVIQ-criterion to classify aphantasia thus leads to the problem as to who might or might not have vivid visual mental images.

An additional concern accompanies self-report questionnaires. Such designs may involve the problem of demand characteristics, according to which not just the motivations of participants (e.g., intention to comply with assumed hypotheses) but also cues from the experimenters (e.g., instructions) could skew experimental findings (Sharpe & Whelton, 2016). Consider the instructions of the VVIQ; first, we are asked to “think of...” a scene or an event, then instructed to “consider carefully *the picture that comes* before your mind’s eye” (Marks, 1973, p. 24, my emphasis). As McKelvie (1995a) pointed out, the latter sentence might indicate to the participant that the experimenter expects a visual mental image, therefore inflating VVIQ scores (at least for non-aphantasic participants). The author thus recommends using a more neutral sentence, such as “Consider *whether* a picture comes before your mind’s eye” (McKelvie, 1995a, p. 83, my emphasis), implying that it is up to the participant’s decision whether a visual mental image is experienced or not. However, McKelvie’s recommendation was not applied to the VVIQ (see e.g., Zeman et al., 2015). Furthermore, Allbutt et al. (2011) presented evidence that there is a correlation between VVIQ scores and social desirability, meaning that high visual mental imagery capabilities are viewed favourably in social contexts. On the other hand, demand characteristics may surface in aphantasia studies in a different manner. Dawes et al. (2020) discussed a potential response bias for aphantasic participants, given that self-identification with a lack of visual mental imagery prior to participating in a study would always imply low scoring on the VVIQ (i.e., an intention to comply with the assumed hypothesis that the study measures a lack of visual mental imagery). This may be a general concern of all studies on aphantasia as these participants are most commonly volunteers, recruited through online platforms created for those who self-describe with aphantasia. On the other hand, Zeman et al. (2015) explicitly stated their expectations towards their 21 volunteering participants that they would rate their “imagery at the low end of the spectrum” (p. 1, in their Supplementary data). Such instructions from researchers could also bias participants towards low scoring on the VVIQ. Whether the low scores of people with aphantasia are due to pre-existing response biases or rather experimental cues from researchers remain unclear.

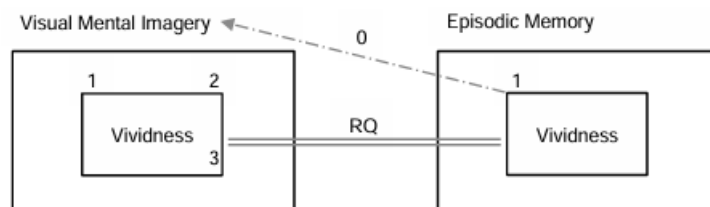
Lastly, as vividness of episodic memory seems to be a parallel research endeavour, a closer inspection of aphantasia’s relation to episodic memory vividness is required. Three main research designs can be distinguished in this literature. First, standardized questionnaires are employed, for example the Survey of Autobiographical Memory (SAM, Dawes et al., 2020) where participants are asked to rate how much they agree with several statements in four categories: episodic, semantic, spatial, and future. Interestingly, it is only the last category that includes explicit references to vividness and mental images (Palombo et al., 2013). Results of Dawes et al. (2020) indicated aphantasic participants having lower scores on the episodic, semantic, and future components of the SAM, compared to control participants. That is, episodic memory is reduced in people with aphantasia, but also visual mental imagery vividness is reduced for future events. There is no indication, however, whether vividness of episodic memory is also reduced. Second, an attempt was made to modify the VVIQ such that it investigates episodic memory vividness. The Episodic Memory Imagery Questionnaire (EMIQ, Dawes et al., 2020; Dawes et al., 2022) instructs participants to remember events and to rate their vividness by utilising the same 5-point rating scale as in the VVIQ and their corresponding labels slightly corrected for recollections (e.g., 1 – *No image at all, I only ‘know’*

that I am recalling the memory). Results showed that aphantasic participants rated lower on the EMIQ than control participants, thus the authors concluded that visual mental images are diminished in episodic memory of aphantasia. The problem with this design, however, is that although participants are asked to recollect past events, there is still reference to mental imagery. As it has been shown earlier, aphantasia generally scores low on the VVIQ (where mental imagery is explicit), therefore it can be assumed that they will rate low on the EMIQ as well. Third, a qualitative approach is employed where, through an autobiographical interview, participants are asked to elaborate on their memories by recollecting details of specific events (Dawes et al., 2022; Milton et al., 2021). The general finding is that participants with aphantasia generate fewer memory details and less vivid recollections than control participants. Thus, the general conclusion is that recollections of aphantasia are diminished, which is correlated with low visual mental imagery capabilities (stemming from low VVIQ scores). It remains unclear, however, whether participants with aphantasia would not interpret memory-vividness instructions as being a task on visual mental imagery, given that the two constructs were measured in two different task designs.

To summarise such concerns, the schematic diagram of Figure 1 shows the concepts and their relationships utilised so far. The main argument of the current thesis states that episodic memory vividness involves visual mental imagery (0). This is based on the following three points. Vividness as a property is used in two theoretical constructs: first, operationalized in visual mental imagery tasks; second, operationalized in episodic memory tasks (1). Yet, theoretical considerations (the depictive view of the mental imagery debate, see before) highlight that vividness is an attribute of visual mental imagery (2). Further to that, vividness is not a clearly defined concept (3). Due to the main argument (0), an additional claim has been made: it is methodologically difficult to separate episodic memory from visual mental imagery.

Figure 1

Schematic diagram of the main argument



Note. 0) Episodic memory vividness involves visual mental imagery. 1) Vividness is operationalized in both visual mental imagery and episodic memory tasks. 2) Vividness is an attribute of visual mental imagery. 3) Vividness is an ill-defined concept. RQ) The main research question.

In other words, the meaning of vividness could be interpreted similarly between “visualizing” and “recalling” a familiar event. As such, the main research question of this thesis is formulated as follows: is there a difference in vividness ratings between visual mental imagery and episodic memory (RQ in Figure 1)? To this end, the current study was designed as follows. To equate task designs of visual mental imagery and episodic memory, two versions of the VVIQ were distributed online among participants; one remaining with the original instructions as in the VVIQ, referring to visual mental imagery, and one modified in a way that all expressions referring to visual mental imagery were substituted by “recollections” or “recall”. Both versions utilised one concept of vividness in their rating scales: clarity. A second research question addresses whether demand characteristics influence vividness rating scores. First, McKelvie's recommendation (1995a) on “consider *whether*...” was implemented in the questionnaires as opposed to the expression “consider *the*...” (my emphases). To further assess social desirability (Allbutt et al., 2011) and the effects of experimental cues, three additional conditions were set up: low, high, and neutral. That is, participants were instructed either that “most people rate low”, “most people rate high” (i.e., on the subsequent questionnaire), or a neutral “no demand characteristic” control condition was employed. Thus, the main hypothesis is that people with or without aphantasia will interpret a memory-vividness task as a measurement of visual mental imagery. Translated to predictions, participants will score similarly on both versions of the questionnaire, while participants self-describing with aphantasia will score lower than control participants. An additional hypothesis is established for demand characteristics; such extraneous variables will affect both aphantasia and non-aphantasia, although in different directions. In terms of predictions, the low demand characteristics condition will affect participants self-describing with aphantasia mostly, that is, resulting in lower scores than in the neutral or high conditions, whereas the high demand characteristics condition will affect participants not self-describing with aphantasia mostly (i.e., the highest scores compared to the other conditions).

Method

Participants

An initial power calculation was performed with G*Power (Faul et al., 2007), with a partial eta squared of .06, derived from a medium effect size $d = .5$, used by Dawes et al. (2020). The recommended sample size was 251 participants (see also in Appendix A). However, literature on aphantasia is inconsistent on the required sample size due to the phenomenon being relatively new (Dawes et al., 2020). Therefore, the goal was to recruit as many participants as possible in the given timeframe of the master thesis project.

Aphantasic participants were recruited through online platforms, such as Facebook groups created for those self-describing with aphantasia: “Aphantasia (Non-Imager / Mental Blindness) Awareness Group”, “Aphantasia Support Group”, “Aphantasia – Survey Group”, and “Aphantasia!”; and a reddit page (r/Aphantasia). Additionally, non-aphantasic participants were also recruited on social media (mostly through Facebook groups) with snowball sampling. One hundred and thirty-nine participants completed the online questionnaire out of which 89 were self-describing with aphantasia (62 females, 25 males, and two “preferring not to say”), whereas 50 were not self-describing with aphantasia (27 females, 22 males, and one “preferring not to say”). Mean age of aphantasic participants was 45.3 years ($SD = 17.4$), whereas 34.5 years ($SD = 12.1$) for non-aphantasic participants. All participants received and read the information letter, after which they provided written consent. The information letter and consent form provided details on the study and the procedure, including voluntary participation and the right to withdraw from the questionnaire without giving a reason, and anonymity during both data handling, analysis, and publication. Given that the study was part of the master thesis, it was approved by the Umeå University.

Materials

The 16 items of the VVIQ were taken from Marks (1973), consisting of four parts: a relative or a friend, a rising sun, the front of a shop, and a country scene. Each part has four items. Each item is to be rated on a 5-point Likert scale where 5 means the highest, whereas 1 means the lowest vividness rating. The maximum total score is 80, whereas the minimum total score is 16. The main instructions for both versions of the questionnaire (details in the next paragraphs) explicated “vividness”, whereas the rating scales included only “clarity” and its different levels: *Perfectly clear and distinct* as the highest rating, whereas *Vague or indistinct* as rating 2. According to McKelvie (1995b), “distinct” is a synonym for “clarity”, whereas “vague” means the opposite. The other interpretations of vividness were omitted in this study: liveliness (Marks, 1973), due to referring to visual qualities that had been reported to be lacking in aphantasia (Keogh & Pearson, 2018); amount of detail, due to being utilised in both visual mental imagery vividness and episodic memory vividness research (D’Angiulli & Reeves, 2007; Geib et al., 2017); and richness (Baddeley & Andrade, 2000), due to being a broad category, which could also include (or imply) amount of detail.

Two versions of the questionnaire were derived from the VVIQ (see in Appendix B) and created by a publicly available online platform: NeuroTask Scripting (NeuroTask B.V., 2021), based on JavaScript. The “visual mental imagery vividness” questionnaire’s wording remained almost the same as the VVIQ (apart from the abovementioned rating scale), that is, it referred to “visual image”, “image”, “picture”, and “mind’s eye”. The instructions for each of the four parts were modified as follows. Participants were asked to “think of” each of the events or scenes, followed by the sentence “consider whether a picture comes before your mind’s eye”. The lowest rating also remained as the original (1 – *No image at all, you only know that you are thinking of the object*).

The “episodic memory vividness” questionnaire’s wording were modified so that all references to mental imagery were replaced by “recall” or “recollection”. The instructions for each of the four parts were altered in a way that participants were asked to “recall” each of the events or scenes, followed by the sentence “consider your experience carefully”. The lowest rating became: 1 – *No recollection at all, you only know that you are thinking of the object*. The main instruction of this version also emphasised that “If you do not have a recollection, rate vividness as ‘1’”. Such a rating description is akin to the original VVIQ’s lowest rating; hence it becomes possible to equate the rating scales of the two modified questionnaires. Although McKelvie (1995a) discusses that a lack of vivid visual mental imagery is analogous to just “thinking” of an event, it does not become directly apparent that a lack of vivid episodic memory also means just “thinking” of it. However, the decision to use such a rating description derives from the study of Reisberg et al. (1988). The authors used a 7-point Likert scale and an additional “does not apply” rating (0 – *events never experienced*), with the lowest rating 1 meaning *events experienced and not remembered*. Such a description implies that there is a case for “thinking” of an event – that had been experienced – although there is no recollection or memory of it during participation.

Procedure

Participants received a URL link that contained one version of the questionnaire. The two manipulations (three demand characteristics: low, neutral, or high; and two questionnaire types: mental imagery or recollection) were allocated pseudo-randomly. That is, upon loading the URL link, a number-generator created an increasing number from one to six, based on which the participant was assigned to a condition. For example, the first condition combined low demand characteristics with the mental imagery questionnaire; the second condition combined low demand characteristics with the recollection questionnaire, and so on, whereas after condition six (high demand characteristics and recollection questionnaire) the number-generator reset to one. In addition, NeuroTask logged every unique participant number by default, thus making it possible to verify during data analysis that none of the participants filled out the questionnaire twice (which was in fact the case). See Table 1 for the number of participants per each condition.

Table 1

Number of participants for each condition (gender in parentheses: female/male, excluding “I prefer not to say”), depending on self-describing with aphantasia or not

Condition (Demand Characteristics & Questionnaire Type)	Self-Describing with Aphantasia	
	Yes (f/m)	No (f/m)
1. Low & Mental Imagery	11 (7/4)	9 (5/4)
2. Low & Recollection	16 (11/4)	9 (3/5)
3. Neutral & Mental Imagery	12 (8/4)	9 (5/4)
4. Neutral & Recollection	13 (11/1)	11 (6/5)
5. High & Mental Imagery	21 (14/7)	4 (3/1)
6. High & Recollection	16 (11/5)	8 (5/3)

Once the URL link was opened, the information letter already provided some details on the “demand characteristics” condition to which the participant was assigned to (but not on the questionnaire type). In the low condition, participants were informed that “Previous research has shown that most people rate low on this questionnaire.”; in the neutral condition the word “low” was substituted by “variously”; whereas in the high condition it was substituted by “high”. After this, participants indicated their consent, which was followed by the actual questionnaire. The main instructions of the questionnaire included those words that indicated the second manipulation (either mental imagery or recollection). Importantly, the demand characteristics condition was repeated (the same sentence as in the information letter) but included an additional sentence. The low condition stated that “most people do not experience” either mental images or recollections; the neutral condition stated that “people either do or do not experience” mental images (or recollections); whereas the high condition stated that “most people do experience” mental images (or recollections). Once participants read the main instructions, they progressed to the 16 items.

At the end of the questionnaire, additional demographic questions were employed, such as the age and gender of the participants, whether they described themselves as having aphantasia, and what device they used to fill out the questionnaire (Desktop/Laptop or Mobile Phone/Tablet). In addition, participants were asked to state what they thought the purpose of the questionnaire was. The duration of the online task took no more than 10 minutes.

Analysis

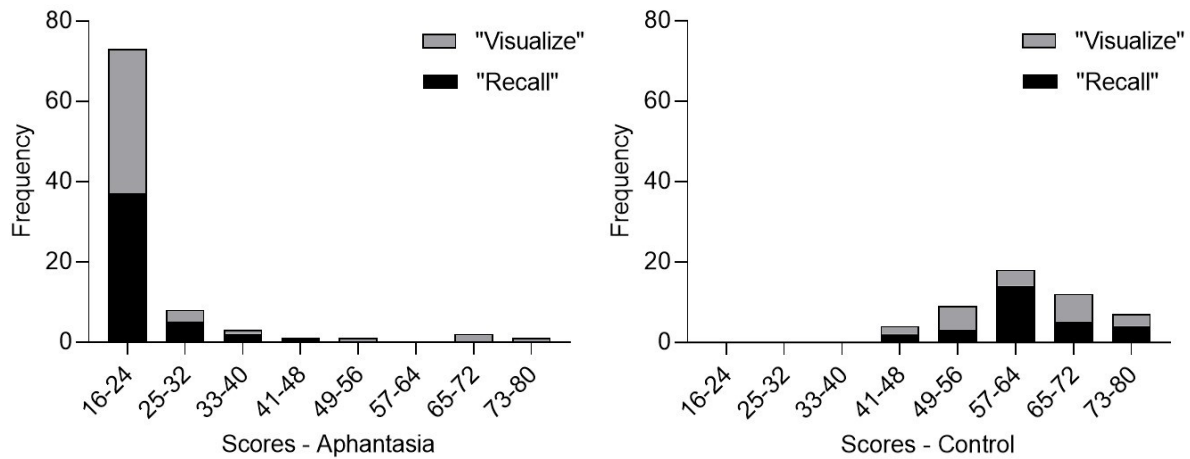
The between-subject design of the analyses included three independent variables and one dependent variable. As for the former one, all three were categorical; group with two levels (aphantasia, control), questionnaire type with two levels (mental imagery, recollection), and demand characteristics with three levels (low, neutral, demand). As for the latter one, responses were measured on a 5-point Likert scale (i.e., five categories, from 1 to 5). Total scores for each participant were collected, which then were averaged over the six conditions (see also in Table 1) separately for the two groups, performed in Microsoft Excel. No data were excluded. As normality assumptions were violated in almost all cases non-parametric tests were performed, apart from analysing participants not self-describing with aphantasia. In addition to this, the distributions of the total scores were dissimilar between the two groups, therefore separate analyses were conducted for each group, as opposed to a three-way ANOVA. First, questionnaire scores were compared between participants with and without self-described aphantasia – irrespective of questionnaire type or demand characteristics – to verify whether self-identification with a lack of visual mental imagery also corresponds to low scores on the questionnaires. Next, the effects of questionnaire types and demand characteristics were analysed, to check whether wording differences (“visualize” or “recall”) or demand characteristics influence questionnaire scores. Thus, the following statistical analyses were conducted: a non-parametric version of an independent samples *T*-test (Mann-Whitney test), a non-parametric version of a two-way ANOVA (Kruskal-Wallis test), and a two-way ANOVA. Statistical significance was determined at a p value $< .05$. Descriptive and statistical analyses were carried out by JASP (JASP Team, 2024), while graphs were created by GraphPad Prism (GraphPad Software, 2023). Finally, checks were made on the closing questions, such as the purpose of the questionnaire and devices used, and on the average time spent on a questionnaire.

Results

For participants self-describing with aphantasia, both Shapiro-Wilk’s test for normality ($p < .001$) and Levene’s test for equality of variances ($p = .034$) were not met. Therefore, a non-parametric Mann-Whitney test was conducted first. This showed that participants self-describing with aphantasia scored significantly lower ($Mdn = 16$, $MAD = 0.0$) than participants not self-describing with aphantasia ($Mdn = 62$, $MAD = 6.5$), $U = 132.50$, $p < .001$, $r_B = -.94$. The median value score of 16 is an indication of no visual mental images at all, which is in line with previous research on aphantasia. However, 10 participants self-describing with aphantasia scored 32 or above; 32 being the average cut-off point of previous studies between good and bad imagers (e.g., Takahashi et al., 2023). There were no participants scoring 32 or below who did not self-describe with aphantasia. See Figure 2 for the frequencies of total scores for each group.

Figure 2

Distribution of the questionnaire scores (total), divided by questionnaire type, for aphantasia and control groups



As for the separate analyses of participants self-describing with aphantasia, a Kruskal-Wallis test showed no significant main effect of questionnaire type, $H(1) = 1.07, p = .301, \eta^2 = .00$. That is, there was no difference between the instruction to “visualize” ($Mdn = 16, MAD = 0.0$) or to “recall” ($Mdn = 17, MAD = 1.0$). This is in line with the prediction that irrespective of how these participants were instructed they still scored low on the questionnaires. Furthermore, there was no significant main effect of demand characteristics on questionnaire scores either, $H(2) = 1.48, p = .477, \eta^2 = -.01$. Table 2 displays further details on each demand characteristics condition. This result is not in line with the current predictions as it shows that irrespective of how these participants were instructed as to what previous research had shown, they still scored similarly. Interaction effects between questionnaire types and demand characteristics were not tested due to the usage of non-parametric tests.

Table 2

Number of participants (n) with self-described aphantasia, median scores (Mdn) and median absolute deviations (MAD) for each demand characteristics condition

Descriptive Statistics	Demand Characteristics		
	Low	Neutral	High
n	27	25	37
Mdn	16	16	16
MAD	0.0	0.0	0.0

As for analysing participants not self-describing with aphantasia, both Shapiro-Wilk’s test for normality ($p = .708$) and Levene’s test for equality of variances ($p = .764$) were met, therefore a two-way ANOVA was conducted. As before, there was no significant main effect of questionnaire type on the questionnaire scores, $F(1, 44) = .07, p = .792, \eta^2 = .00$. This result again corroborates earlier predictions, meaning that irrespective of whether these participants were instructed to “visualize” ($M = 61.9, SD = 10.6$) or to “recall” ($M = 62.8, SD = 8.3$) they scored similarly on the questionnaires. Furthermore, there was a significant main effect of demand characteristics on the questionnaire scores, $F(2, 44) = 6.12, p = .005, \eta^2 = .21$. As predictions on demand characteristics involved comparing each level of the independent

variable with subsequent levels, Helmert contrasts were employed. These revealed that the score-differences between the high and low/neutral conditions were not significant, $t(44) = .94$, $p = .351$, whereas the score-differences between the low and neutral conditions were significant, $t(44) = -3.39$, $p = .001$. Table 3 shows further details on each demand characteristics condition. This would indicate that – contrary to predictions – only the low demand characteristics condition affected questionnaire scores for those who do not self-describe with aphantasia. Furthermore, there was no interaction effect between questionnaire type and demand characteristics, $F(2, 44) = 1.53$, $p = .229$, $\eta^2 = .05$. Figure 3 displays the distribution of total scores of each demand characteristics condition.

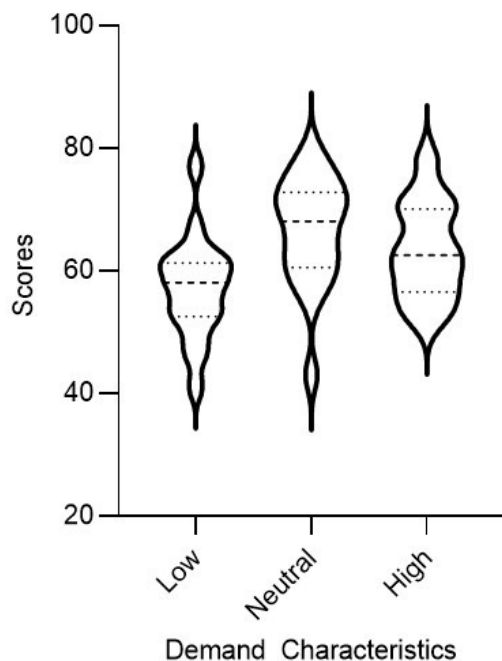
Table 3

Number of participants (n) without self-described aphantasia, mean scores (M) and standard deviations (SD) for each demand characteristics condition

Descriptive Statistics	Demand Characteristics		
	Low	Neutral	High
n	18	20	12
M	57.1	66.6	63.3
SD	8.2	8.9	7.9

Figure 3

Violin plot of the distribution of total scores of each demand characteristics condition for participants not self-describing with aphantasia, with medians and first and third quartiles



In addition, 20 out of the 139 participants indicated “incorrectly” what the purpose of the questionnaire was, out of which 12 were self-describing with aphantasia. That is, while these participants received instructions to “recall” the items (i.e., no indication of visual mental imagery) they still thought that the questionnaire was about either mental imagery (or variations, such as “visualization”, the “mind’s eye”, or “imagination”) or that it was about

finding out how many people have aphantasia. Furthermore, this was only true in case of the “recall” conditions (2, 4, and 6 in Table 1); those who received instructions to “visualize” the items did not indicate that the questionnaire was about memory. Although no statistical analysis was performed on these free texts, such comments strengthen the initial argument that episodic memory vividness tasks involve visual mental imagery.

Finally, a glance at the devices used by the current study’s participants revealed that 76 people filled out a questionnaire on mobile phones (or tablets), whereas 62 people utilised a desktop (or a laptop)⁴. Additionally, participants self-describing with aphantasia spent five minutes, whereas control participants spent seven minutes – on average – on filling out a questionnaire.

Discussion

The current study aimed at investigating how one theoretical construct (episodic memory) is involved in a task with another theoretical construct (visual mental imagery), making an implicit connection between the two by the concept of vividness. As it was predicted, neither participants self-describing with aphantasia nor control participants scored differently in the two versions of the VVIQ (Marks, 1973). That is, irrespective of whether participants were instructed to “visualize” or to “recall” several scenes, item scores were comparable on the questionnaires. The low scores of participants self-describing with aphantasia on both questionnaires further strengthen the study’s initial claim since this population is described by an absence of voluntary visual mental imagery (Zeman et al., 2015). A secondary goal of the study was to investigate demand characteristic effects on these questionnaires, that is, whether participants’ motivations or cues from researchers could alter task performance (Sharpe & Whelton, 2016). The predictions derived from experimental cues were not met. Scores of participants self-describing with aphantasia were not influenced by the initial statements on alleged findings of previous research (i.e., low, neutral, or high conditions). Surprisingly, the results of control participants displayed the opposite direction as it was expected. Whereas Allbutt et al. (2011) showed that high visual mental imagery capacities are favoured in the general population, the current manipulation of the high demand characteristic condition did not yield a significant difference from the other conditions. In fact, the instruction that “most people rate low” on the questionnaire resulted in the lowest scores for control participants, compared to the other two conditions. This unexpected result cannot be explained following earlier considerations on demand characteristic effects in the general population, thus further research would be required.

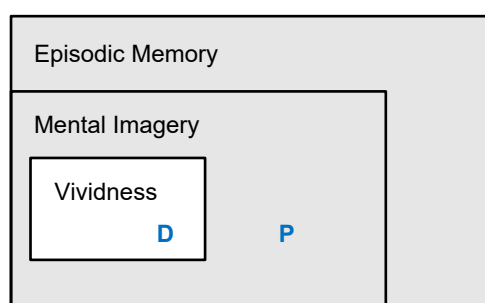
The basis of this study was formulated around the following premises, paraphrased from the main argument (see also in Figure 1): vividness is a property of visual mental imagery (e.g., Marks, 1973; McKelvie, 1995a); vividness is not a clearly defined concept (e.g., Fazekas et al., 2020; Schwitzgebel, 2002); and vividness is also associated with episodic memory (e.g., Cooper & Ritchey, 2022; Geib et al., 2017). It was hypothesized that a task on memory-vividness will be interpreted as a measurement of visual mental imagery irrespective of aphantasia or not. The current results favour this hypothesis. Such an interpretation is further derived from the fact that the items of the VVIQ stem from familiar scenes and events that are easy to recall (Marks, 1973). The main argument and its derived hypothesis offer implications on the mental imagery debate as well. Figure 4 proposes a potential candidate on how to conceptualize the relationships between episodic memory, visual mental imagery, and vividness. Accordingly, mental imagery is part of episodic memory. Vividness is associated with (or part of) episodic memory. However, this will involve visual mental imagery since vividness is a property of mental imagery. Applying this conceptualization to the mental imagery debate, the depictive format of mental imagery (Pearson & Kosslyn, 2015) could be placed in the most inner rectangle of Figure 4 (in Vividness), since definitions of vividness

⁴ With one participant leaving the field empty.

refer to perceptual qualities. Furthermore, the propositional format (Pylyshyn, 1979) could be placed inside the Mental Imagery rectangle (but outside Vividness), meaning that a propositional format of mental imagery is admissible next to a depictive format, as opposed to taking one stance in the mental imagery debate (Pearson & Kosslyn, 2015). Understood in this way, it could be the case that the general population could have depictive mental imagery (or perhaps, even propositional). Further to that, people with aphantasia might have only propositional imagery. Figure 4 also shows that there is a case for episodic memory without mental imagery, that is, episodic memory could also be propositional. Whether such a theoretical conceptualization (an extension of the mental imagery debate) offers testable hypotheses remains open for future research. This study rather emphasises the methodological difficulty of disentangling two separate theoretical constructs and cautions for the hardship of making conclusions on the “wrong” one.

Figure 4

Schematic diagram of the proposed relationships between the theoretical constructs



Note. White background refers to a depictive, whereas the grey background refers to a propositional representation. D) Depictive mental imagery. P) Propositional mental imagery.

In line with this interpretation, it is important to emphasise alternative explanations or conclusions as well. Dawes et al. (2020) suggested earlier that self-identifying with aphantasia prior to a study would always result in low scoring on such tasks. Since the instructional cues employed in this study did not manage to affect the questionnaire scores of aphantasic participants, the idea of such a bias poses some interesting questions. Presume that the response bias exists; would that mean that participants self-describing with aphantasia scored low on both questionnaire types (“recall” or “visualize”) *solely* because of that? Does the possibility of such a response bias exclude the initial hypothesis that a measurement of memory-vividness is interpreted as a measurement of visual mental imagery? In other words, scoring similarly low on both questionnaire types does not necessarily mean that the task was interpreted such and such way, but perhaps that these participants had already identified themselves as having aphantasia. However, the possibility of such a response bias in aphantasia cannot explain why control participants scored comparably on both questionnaire types as well.

Another conclusion can be derived from earlier studies. Previous research has shown that participants with aphantasia have reduced episodic memory (Dawes et al., 2020), impaired visual mental imagery in episodic memory (Dawes et al., 2020; Dawes et al., 2022), or less vivid autobiographical memories (Dawes et al., 2022; Milton et al., 2021), compared to control participants. In line with these conclusions, it could be argued that the current study’s “visualize” questionnaire measured visual mental imagery and the “recall” version measured episodic memory. Hence, arguing that people with aphantasia have both diminished visual mental imagery and episodic memory, due to scoring low on both questionnaires. However, such an interpretation – based on the methodology employed in this study – is problematic. A closer inspection of the low scores of participants self-describing with aphantasia (*Mdn* = 16 on “visualize” and *Mdn* = 17 on “recall”, i.e., scoring low on all or almost all of the 16 items) and the corresponding rating scale descriptions would indicate that these participants have “no

image at all, they only know that they are thinking of the object” and have “no *recollection* at all, they only know that they are thinking of the object”. Given that the label *aphantasia* – “a condition of reduced or *absent* voluntary *imagery*” (Zeman et al., 2015, p. 379, my emphasis) – was coined after participants reached a median score of 16 on the VVIQ, the current results would imply that *aphantasia* also means a condition of reduced or *absent* voluntary *memory*. This, however, cannot be the case as research on *aphantasia* has only demonstrated reduced or a lack of visual mental imagery and a reduced (but not a lack of) episodic memory in this population. Therefore, the original interpretation would be more applicable in the current study: an episodic memory vividness task involves visual mental imagery as well.

There are at least three limitations surrounding the methodology of the current study. As pointed out earlier, vividness is a problematic concept. It is described interchangeably by liveliness, clarity, amount of detail, richness, or brightness (e.g., D’Angiulli & Reeves, 2007; Marks, 1973; Schwitzgebel, 2002). While the current study employed only one “definition” of vividness (i.e., clarity), a claim cannot be made that this is the “right definition”. Since vividness was not manipulated, for example, by utilising one description in one version of the questionnaire and another one in the second one, possible scoring differences between concepts of vividness were not investigated. This also highlights the problem of a lack of objective vividness-criterion, which could afford proper comparisons between vividness ratings (McKelvie, 1995a; Schwitzgebel, 2002). Second, the rating scale employed in this study followed the traditional structure of a 5-point Likert scale that is used in subjective measurements of visual mental imagery (e.g., Marks, 1973; Zeman et al., 2015). McKelvie (1995a), however, recommended that a 7-point Likert scale would be more suitable for such purposes. Given that *aphantasia* is described by a diminished or a lack of visual mental imagery and by a diminished (but not a lack of) episodic memory, it might be expected that a “visualize” questionnaire would yield significantly lower scores than a “recall” questionnaire, contrary to the initial predictions. However, it could be argued that rating such experiences on five points does not give much room to perfectly discriminate between the two constructs. By employing, for example, seven points, differences in visual mental imagery and episodic memory could potentially become more discernible. Perhaps, participants self-describing with *aphantasia* might score 3 or 4 on each item of the “recall” questionnaire, whereas 1 or 2 on each item of the “visualize” questionnaire. If this was the case, the argument that episodic memory vividness involves visual mental imagery would be in peril.

A third limitation derives from an uneven distribution of participants in the employed six conditions (see also in Table 1). Although participants were quasi-randomly assigned to each condition, achieved by a sequential number-generator created in NeuroTask, the six conditions did not result in equal amounts of participants. For example, there were 21 participants self-describing with *aphantasia*, whereas only four control participants in the fifth condition (high demand characteristics and mental imagery questionnaire). The low number of participants in some of the conditions could be one reason why the predicted effects of demand characteristics were not found. This might have been circumvented by a continuous tracking of participant-assignment. However, due to the nature of the online recruitment (i.e., the study was posted on multiple platforms and shared among many people) it was not possible to manually monitor proper allocation of the conditions. Further to that, gender distribution was uneven between the conditions. For example, only one male but 11 females self-describing with *aphantasia* were in the fourth condition (neutral demand characteristics and recollection questionnaire). In fact, female participants were overrepresented in those self-describing with *aphantasia* (62 as opposed to 25 males). Why this was the case was not investigated in the current study. A solution would be to match gender between *aphantastic* and control participants (e.g., in Dawes et al., 2022).

Some directions for future research can also be laid out. On the one hand, it would be necessary to replicate the study in a controlled laboratory environment where participants are seated in an undisturbed room and hence could focus more on their inner experiences while

considering the items of the questionnaire. As results indicated, more participants used mobile phones (or tablets) than desktops (or laptops). It could be argued that filling out a questionnaire on a mobile phone's screen might be less engaging than being seated alone in a dimly lit room and in front of a computer screen. On the other hand, although the current study did not aim at measuring the time spent on a questionnaire, NeuroTask recorded the exact times of opening and closing the URL link, showing an average of five to seven minutes of online participation. Partaking physically in a study could possibly involve more engagement and more time to introspect on such subjective experiences.

A further suggestion is to expand the scope of modified instructions to other cognitive tasks that aim to measure visual mental imagery (or episodic memory). Pearson et al. (2013), for example, provide a general framework for experimental paradigms on visual mental imagery. Previous research has shown that behavioural performances on such objective (as opposed to a subjective questionnaire) tasks are comparable between aphantasic and control participants, for example, in animal tails tests or straight/curved letter tasks (Milton et al., 2021) or in mental rotation tasks (Milton et al., 2021; Pounder et al., 2022). This suggests that people with aphantasia utilise an alternative strategy to visual mental imagery (Liu & Bartolomeo, 2023). Investigating alternative cognitive strategies could become also possible by implementing different instructions, for example, by either “recalling” or “visualizing” varying animals or letters or even the three-dimensional objects of the mental rotation task. The requirement for such objective measurements is further necessitated by the fact that there is no current objective criterion to identify aphantasia. As previous research has shown there is no agreed cut-off point on the VVIQ to distinguish aphantasics from control participants (e.g., Takahashi et al., 2023; Zeman et al., 2020), which also became apparent in the current study by those 10 participants who, although self-described with aphantasia, still scored 32 or above. In addition to the initial argument of this study, it could also be argued – albeit with caution – that the empirical design employed here aimed at offering a novel approach as to how to measure a (lack of) visual mental imagery without explicitly referring to it.

In summary, this study demonstrated that vividness of visual mental images and of recollections can be interpreted as similar experiences, and that it is methodologically difficult to separate the two theoretical constructs. From a participant's perspective, it can be the case that visualizing or recalling a friend's face can be subjectively very similar. If vividness is not a clearly defined empirical concept, can it not be assumed that it is also used interchangeably in everyday expressions? When I visualize a friend's face, it proves to be hard for me to discern whether its vividness is part of my mental image or of my memory. What the case is in the experience of people with aphantasia might be proven to be even harder. Such intuitive concerns, however, should be used as premises to further our empirical curiosity and to find better methodological ways to investigate mental imagery, episodic memory, and vividness.

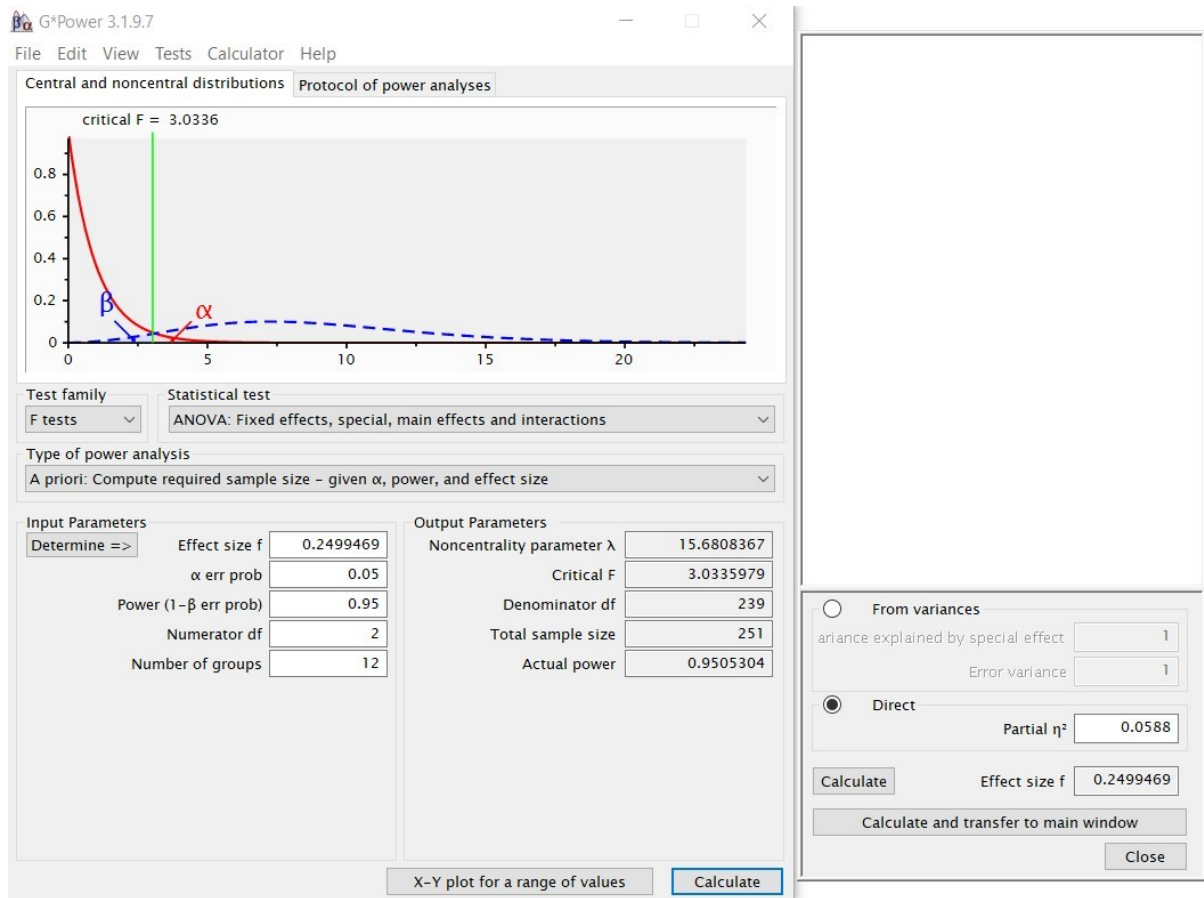
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Appendix A

G*Power Calculation



Appendix B

Questionnaire – Visual Mental Imagery

For each item on this questionnaire, try to form a visual image, and consider your experience carefully. For any image that you do experience, rate how vivid it is using the five-point scale described below. If you do not have a visual image, rate vividness as '1'. Only use '5' for images that are truly vivid. Please note that there are no right or wrong answers to the questions, and that it is not necessarily desirable to experience imagery or, if you do, to have more vivid imagery.

Perfectly clear and distinct	5
Clear	4
Moderately clear	3
Vague or indistinct	2
No image at all, you only know that you are thinking of the object	1

For items 1-4, think of some relative or friend whom you frequently see (but who is not with you at present) and consider whether a picture comes before your mind's eye.

1. The exact contour of face, head, shoulders and body _____
2. Characteristic poses of head, attitudes of body etc. _____
3. The precise carriage, length of step etc., in walking _____
4. The different colours worn in some familiar clothes _____

For items 5-8, think of the rising sun. Consider whether a picture comes before your mind's eye.

5. The sun rising above the horizon into a hazy sky _____
6. The sky clears and surrounds the sun with blueness _____
7. Clouds. A storm blows up with flashes of lightning _____
8. A rainbow appears _____

For items 9-12, think of the front of a shop which you often go to. Consider whether a picture comes before your mind's eye.

9. The overall appearance of the shop from the opposite side
of the road _____
10. A window display including colours, shapes and details
Of individual items for sale _____
11. You are near the entrance. The colour, shape and
details of the door. _____
12. You enter the shop and go to the counter. The counter
Assistant serves you. Money changes hands _____

Finally, for items 13-16, think of a country scene which involves trees, mountains and a lake.
Consider whether a picture comes before your mind's eye.

13. The contours of the landscape _____
14. The colour and shape of the trees _____
15. the colour and shape of the lake _____
16. A strong wind blows on the trees and on the lake causing
reflections in the water. _____

Questionnaire – Episodic Memory

For each item on this questionnaire, try to recall a scene or an event, and consider your experience carefully. For any recollection that you do experience, rate how vivid it is using the five-point scale described below. If you do not have a recollection, rate vividness as '1'. Only use '5' for recollections that are truly vivid. Please note that there are no right or wrong answers to the questions, and that it is not necessarily desirable to experience a recollection or, if you do, to have more vivid recollections.

Perfectly clear and distinct	5
Clear	4
Moderately clear	3
Vague or indistinct	2
No recollection at all, you only know that you are thinking of the object	1

For items 1-4, recall some relative or friend whom you frequently see (but who is not with you at present) and consider your experience carefully.

1. The exact contour of face, head, shoulders and body _____
2. Characteristic poses of head, attitudes of body etc. _____
3. The precise carriage, length of step etc., in walking _____
4. The different colours worn in some familiar clothes _____

For items 5-8, recall the rising sun. Consider your experience carefully.

5. The sun rising above the horizon into a hazy sky _____
6. The sky clears and surrounds the sun with blueness _____
7. Clouds. A storm blows up with flashes of lightning _____
8. A rainbow appears _____

For items 9-12, recall the front of a shop which you often go to. Consider your experience carefully.

9. The overall appearance of the shop from the opposite side
of the road _____
10. A window display including colours, shapes and details
Of individual items for sale _____
11. You are near the entrance. The colour, shape and
details of the door. _____
12. You enter the shop and go to the counter. The counter
Assistant serves you. Money changes hands _____

Finally, for items 13-16, recall a country scene which involves trees, mountains and a lake. Consider your experience carefully.

13. The contours of the landscape _____
14. The colour and shape of the trees _____
15. the colour and shape of the lake _____
16. A strong wind blows on the trees and on the lake causing
reflections in the water. _____